PART B — HEALTHCARE PROBLEM STATEMENTS

H1. OPD Triage Assistant for an Overloaded Government Hospital

Real-world pain: An outpatient department sees 800 walk-ins a day with two doctors. Patients are seen first-come-first-served, so a person with chest pain waits behind 40 people with mild fever. The triage decision today is made by an overworked nurse in 30 seconds.

The problem: From a short intake form and a free-text complaint, assign a triage priority and recommend the right department — safely.

What we provide:

- 30k synthetic intake records: age, sex, vitals (BP, pulse, temp, SpO2), free-text chief complaint, known conditions, medication list
- Complaint text written the way patients actually speak, translated/transliterated across languages
- A published 5-level triage scale with clinical definitions (an ESI-style rubric) in the domain primer
- Hidden labels: correct triage level and correct department, plus a flag for the ~4% of cases that are true emergencies

Assumptions students may make:

- No labs, imaging, or ECG available at triage — only the intake form
- Vitals may be missing or mis-entered (teams must handle 0 BP, temp of 96°C)
- Patients may under-report or over-report symptoms
- Nurse has ≤60 seconds per patient, so the interface must be near-instant

Desired output:

1. A triage level (1-5), recommended department, and the **red-flag symptoms detected**
2. **Asymmetric safety is the core requirement:** under-triaging a true emergency must be penalised ~20× more than over-triaging a mild case. `evaluate.py` implements this — teams that optimise plain accuracy will score badly, and discovering this is part of the learning
3. A mandatory hard-coded escalation rule layer that overrides the model on critical vitals (teams must show these rules)
4. Sensitivity/recall on the hidden true-emergency set, reported separately and prominently
5. An abstention mechanism: the system may say "cannot assess — send to nurse now" and is rewarded for using it appropriately

Stretch: estimate expected wait-time reduction versus the current first-come-first-served baseline using the provided arrival log.